

ASIX: An Analog Spectral Imager for X-rays

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Main goal and motivation

Develop a new class of hybrid X-ray event-driven detector for imaging and spectroscopy, high throughput applications (mainly space, with potential in other fields).

Design goals:

- High spatial ($<10\ \mu\text{m}$) and spectral (350 eV FWHM @ 8 keV) resolution
- Maximum global readout rate: 100kHz

Integrated Front-end electronics:

- Detect and process analog signals from single photons, obtained in real time.
- Room temperature operation: 1–50 keV energy range
- Prototypes qualified at highest TRL possible (TRL-6)

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Background

Frame-driven Readout

- Full-frame coverage, **spatial resolution** set by pixel pitch and array size.
- Integrated per frame. **Energy resolution** performance limited by pile-up and saturation.
- **Low Throughput: limited by pile-up** and full-array monolithic readout.
- **Analog readout.**

Pixel-driven Readout

- Random or selective access to individual pixels. No sub-pixel improvement.
- **Limited by charge sharing through threshold effects:** Hard to get very good energy resolution
- Selectively read sub-regions with higher effective rate: Trade spatial coverage for local speed.

Event-driven Readout

- **Spatial resolution** set by pixel pitch. **Support sub-pixel resolution** via charge-sharing/event processing.
- **High spectroscopic fidelity:** Per-event energy counts.
- **Immune to pile-up.**
- **Mitigates threshold effects.**
- **Moderate throughput:** Hard to get very high rates.
- **Analog readout.**

Team track record: Design, development and integration of advanced detectors and detection systems and their qualification for use in space (Imaging X-ray Polarimetry Explorer - IXPE)

Project Impact

System: single photons fully processed in an event-driven fashion. Photon information (position, energy, arrival time) available at the same time.

- **Immune to pile up and multiple counting**
- **Spatial resolution not limited by charge sharing:** sub-pixel resolution via charge barycentering
- High spectral resolution + high-resolution imaging over large areas.
- Maximum throughput achievable boosted by the trigger and event size selection logic embedded.

Fully resolving the Cu-K_α and Cu-K_β lines.

Applications in X-ray telescopes, satellite remote sensing and material science research. ASIX high energy resolution enables sharp energy windowing.

Higher Spatial Resolution

- Smaller pixel pitch
- Low energy thresholds
- Charge-sharing / centroiding
- Event-driven readout

Trade-offs

Higher Energy Resolution

- Larger pixel size
- Low noise
- No charge sharing

Pixel pitch

Threshold level

Sensor thickness

Parallellism:
Speed-up
readout

Self-triggering
logic

High spatial
resolution
($<10\ \mu\text{m}$)

50 μm pixels in
hexagonal array

Single photon,
ultra-low-noise
analog readout
($< 30\ \text{e}^-$ ENC)

High energy
resolution
(350 eV FWHM at
8 keV)

Prototype

Built upon:

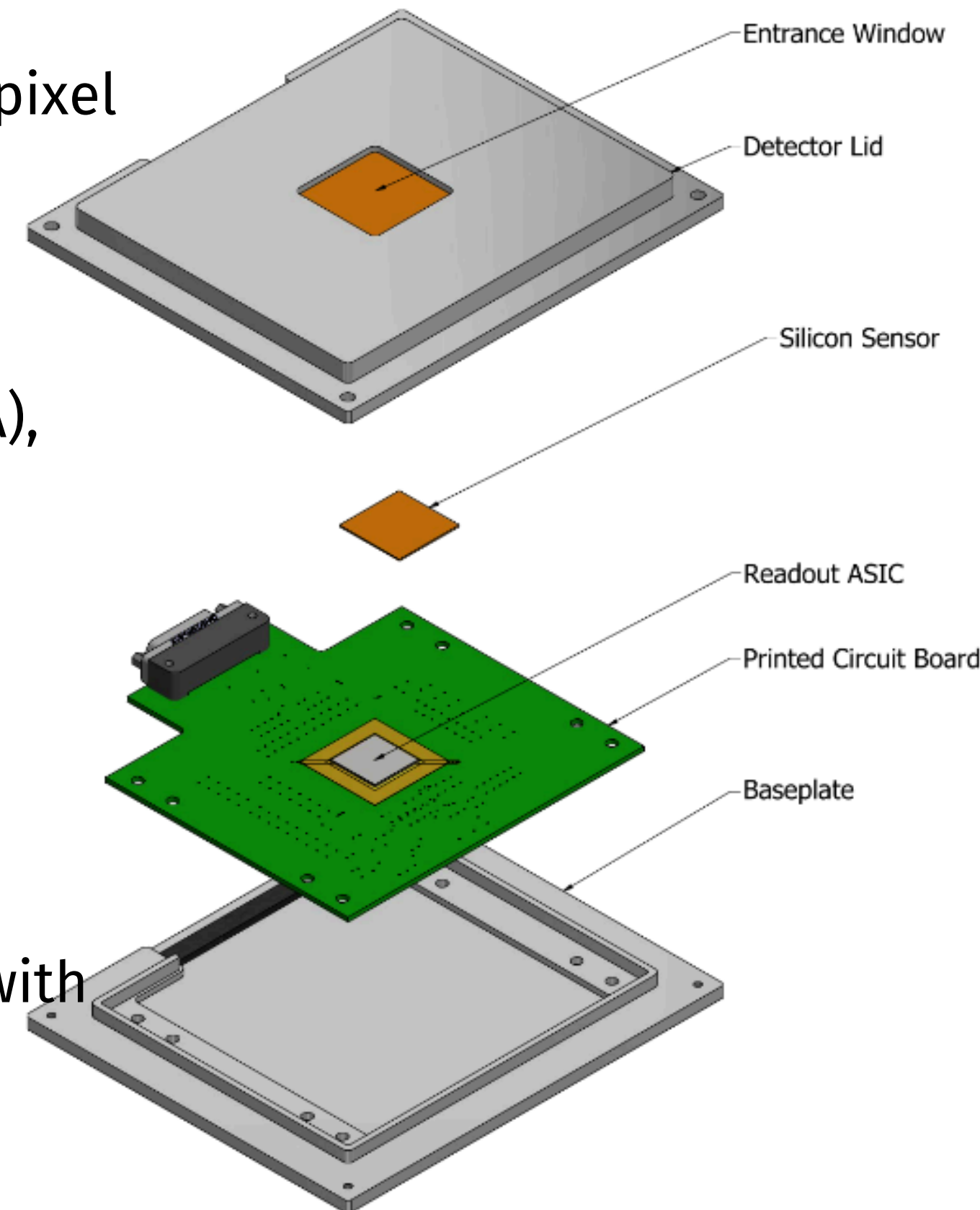
- XPOL-III (*Minuti et al, 2023*): Large scale ($\sim 2\text{cm}^2$) readout ASIC (pixel hexagonal matrix).
- FPGA based Data Acquisition system (DAQ) adapted from IXPE
- $50\text{ }\mu\text{m}$ pixels in hexagonal array
- Low-noise spectroscopic chain: Charge-Sensing Amplifier (CSA), shaper and self-triggering peak detector circuit.

Charge detection $\rightarrow$ ROI isolation $\rightarrow$ Readout sequence:

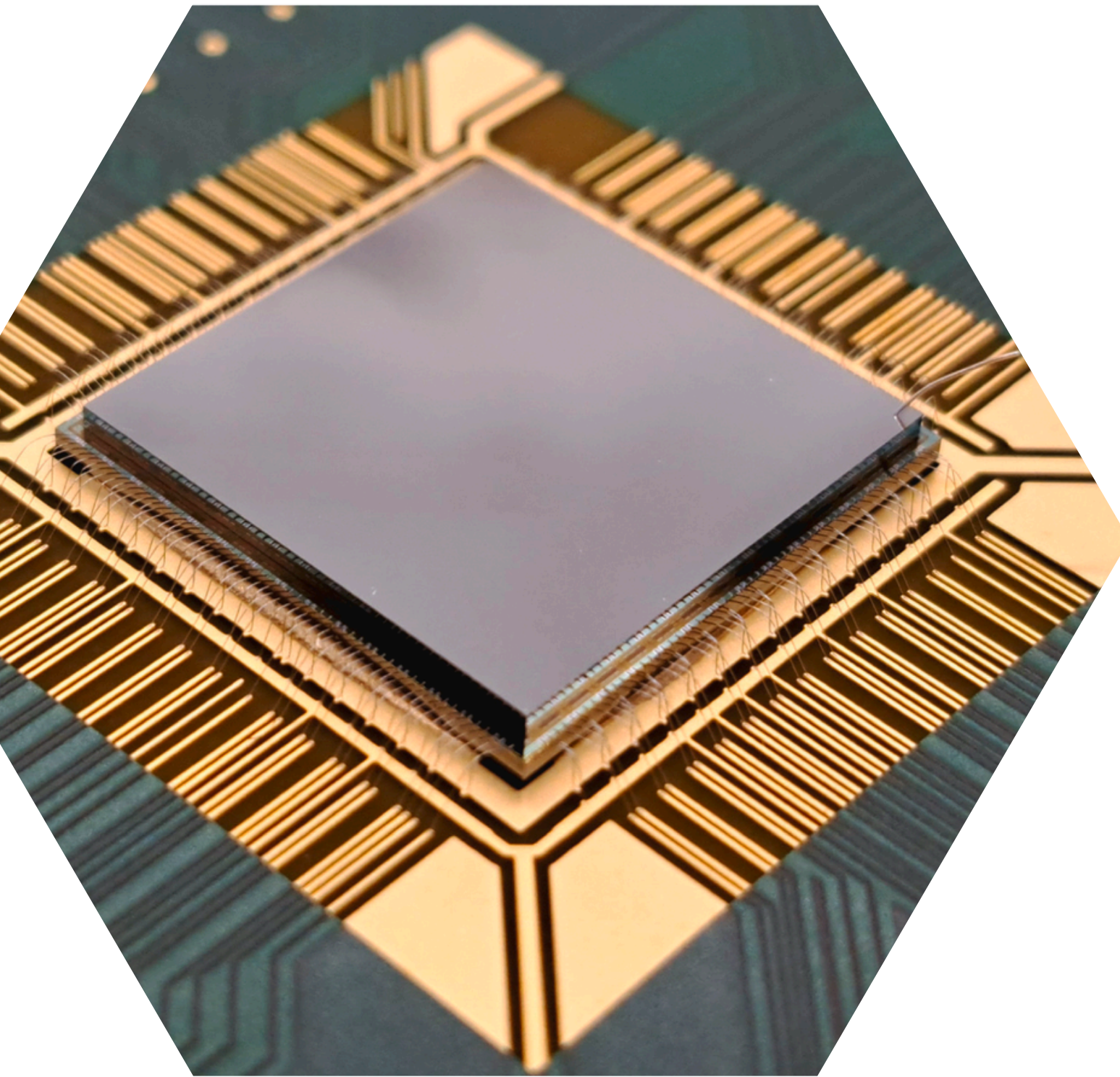
- Off-chip ADC $\rightarrow$ precise single-photon processing.

Two HPD sensor variants:

- $750\mu\text{m}$ thick Cadmium-Telluride (CdTe) Schottky-type sensor with $100\mu\text{m}$ pixels.
- $300\mu\text{m}$ thick n-on-p Silicon (Si) sensor with $50\mu\text{m}$ pixels.



Prototype: Assembly and Initial test



Hybrid Pixel Detector with CdTe sensor.
Taken from: Minuti, M. et al (2026)

Installation in a custom designed detector housing:

- Mylar foil: light shielding
- dry-air environment → Allow to cool down to $-30\text{ }^{\circ}\text{C}$ in order to study the performance as a function of the temperature.

Evaluation of Pixel noise:

- 60 e^{-} ENC for CdTe Vs. 70 e^{-} ENC for Si.
- **Sensor leakage current negligible:** No significant change in noise across a $50\text{ }^{\circ}\text{C}$ operating temperature range
- Overall noise performance within $\pm 10\%$ of our original expectations

Prototype: Spatial Resolution

Huttner 18-lead test pattern image response + Edge-Spread-Function (ESF) analysis.

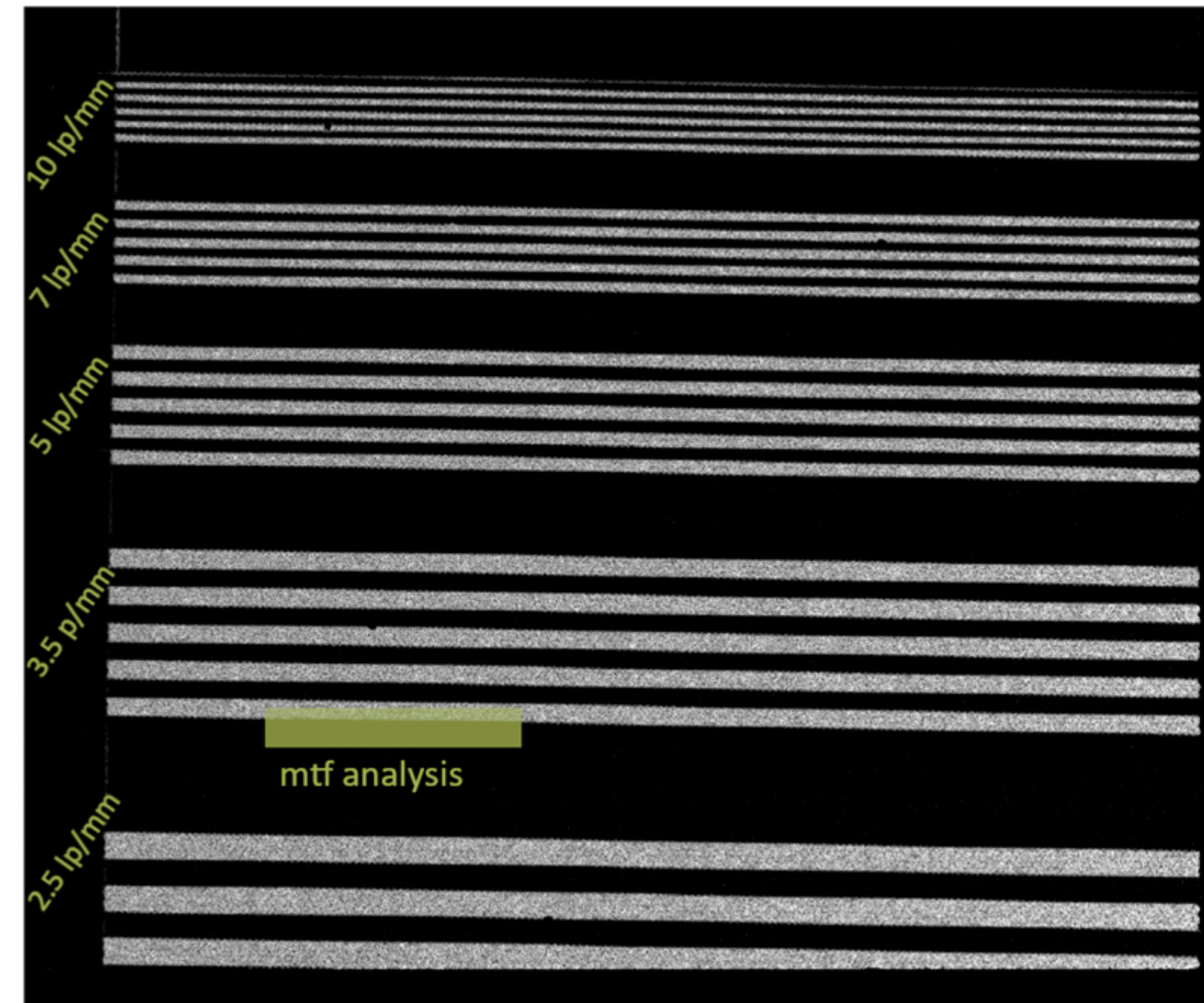
Illumination:

- CdTe: Silver (Ag) target X-ray tube @ 25 kV.
- Si: Gold (Au) target X-ray tube @ 10 kV.

Imaging performance evaluation:

- Binning the estimated photon absorption point onto a two-dimensional reticle.
 - (CdTe: $15 \times 15 \mu\text{m}$; Si: $10 \times 10 \mu\text{m}$ pixel).
- Events selected with up-to four pixels ($> 90\%$ of the total).
- Absorption point computed by basic center-of-gravity calculation.

Spatial resolution: CdTe $\sim 20 \mu\text{m}$, Si $\sim 7 \mu\text{m}$.



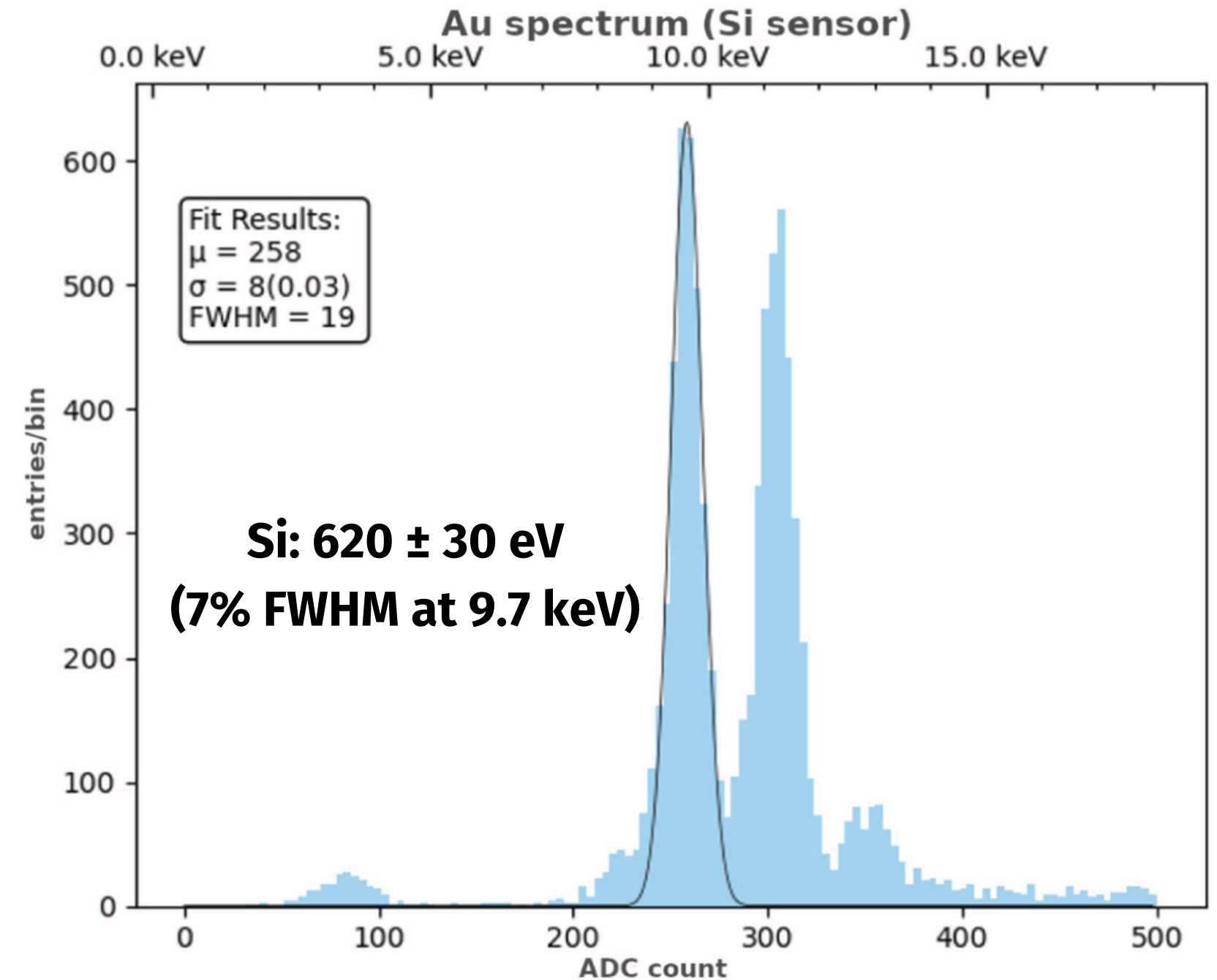
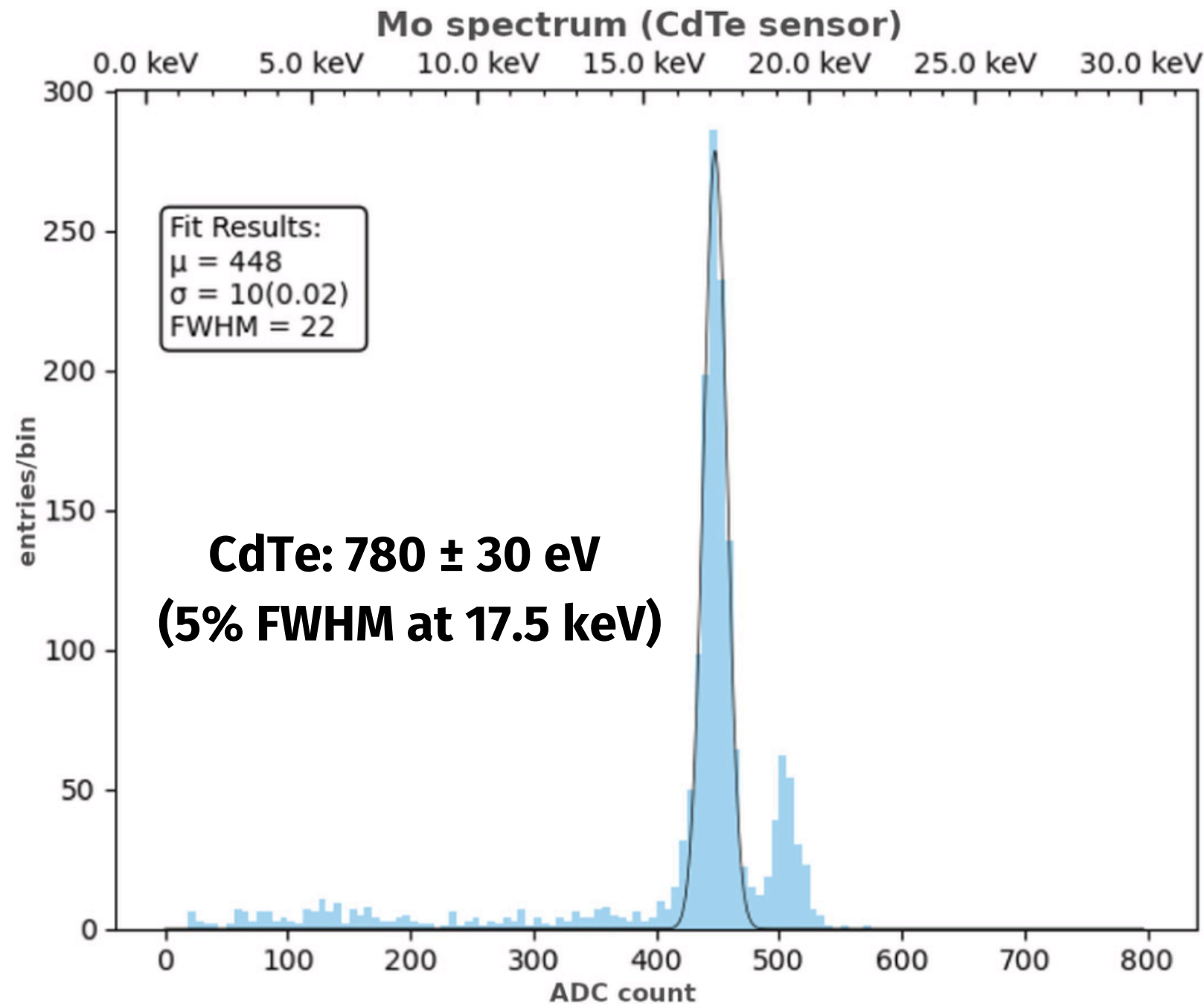
Huttner test pattern image (left) recorded by the silicon device for the evaluation of the spatial resolution.

Taken from: Minuti, M. et al (2025)

Prototype: Energy Resolution

Analyzing the characteristic fluorescence radiation emitted by high-purity samples.

- CdTe sensor: Mo K-lines fluorescence emitted by a 99.9% pure Mo sample.
- Si sensor: Au L-lines fluorescence spectrum emitted by a 99.9% pure Au sample.



MVPs spectral data. Taken from: Minuti, M. et al (2026)

Prototype: Summary

Parameter	CdTe	Si
Number of pixels	26,752 (152×176)	107,008 (304×352)
Pixel size	100 μm	50 μm
Thickness	750 μm	275 μm
Shaping time	1 μs	1 μs
Pixel Noise (ENC)	60 e^-	70 e^-
Energy resolution	800 eV (Mo- K_α)	660 eV (Au- L_α)
Spatial resolution ^a	20 μm	7 μm
Full scale linear range (FSLR)	130 keV	110 keV
Minimum trigger threshold	~ 1.5 keV	~ 1.0 keV

^a X-ray source: X-ray, Ag (25 kV) for CdTe, Au (10 kV) for Si.

Summary of the basic properties of the ASIX prototype HPDs.
Taken from: Minuti, M. et al (2026)

Conclusions and Discussion

ASIX's outstanding features: Hybrid pixel detectors versatility, fine pitch pixels, fully-analog event-driven readout, targeted energy and spatial resolution.

Energy Resolution: For the Si sensor, the spread in the energy measurement is primarily dominated by the electronics noise. *Readout chip was not designed to have a sensor attached.*

Prototype performance: Initial step towards the feasibility of achieving simultaneous high spatial and energy resolution using a small-pixel HPDs.

Next step: Develop new sensors based on a series of freshly designed readout ASICs.

- Core readout principles and pixel geometry inherited from XPOL-III.
- Faster (parallel) readout, reduced electronic noise: Higher energy resolution.

This class of analog readout devices will be highly competitive candidates for applications requiring very precise X-ray imaging and energy discrimination

References

- 1 - Minuti, M. et al. (2026)** *“Towards high-resolution X-ray spectral imaging”*. Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, Vol. 1082, Part 2. doi: 10.1016/j.nima.2025.171001
- 2 - Minuti, M et al. (2025)** *“ASIX: Single-photon, energy resolved X-ray imaging with 50 μm hexagonal hybrid pixel”*. Front. Sens. 6:1599365. doi: 10.3389/fsens.2025.1599365
- 3 - Minuti, M et al. (2023)** *“XPOL-III: A new-generation VLSI CMOS ASIC for high-throughput X-ray polarimetry”*. Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, Vol. 1046. doi: 10.1016/j.nima.2022.167674



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